INVESTIGATION OF THE DYNAMICS OF TRIPLE SYSTEMS BY THE METHOD OF STATISTICAL TESTS. II

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Astrofizika, Vol. 4, No. 1, pp. 31–40, 1968

The equations of motion for a triple system have been integrated numerically on a computer for 100 random initial configurations. The components of the system were assumed to be at rest at the initial time. The equations were linearized in the case of close binary approaches of the components. In all the systems the motions terminated in decay. In each case the decay was a consequence of a close triple approach of the components. The mean decay time $\bar{T}$ was found to be to be 110.1 τ, where τ is the mean time of the component motion through the system. The root-mean-square deviation σ of the decay time from the mean is 125.0 τ.

By linearizing the equations of motion we have been able to fill the gap in the calculations reported in [1] for 38 initial conditions with small velocity of the components at the initial time. For these cases the calculations were carried out up to the decay stage. From the data reported here and the data in [1] we found that $\bar{T} = 80.7\,\tau$ and $\sigma = 83.0\,\tau$ for 100 systems with random initial configurations and small initial velocities of the components. For a set of 200 triple systems with random initial conditions we found that $\bar{T} = (95.4 \pm 6.9)\tau$.

A study of the dynamics of triple systems by integration on a computer of the equations of motion was reported in [1] for the plane problem and different randomly specified initial conditions.

Since the equations of motion were not linearized in [1] the computer could not complete the calculations whenever two of the components of the system underwent a very close approach. The most interesting case, i.e., when all three components of the system were at rest at the initial time, could not therefore be investigated. In this particular case, a very close binary approach of the components occurs almost immediately after the beginning of the motion, and integration cannot be performed. To avoid this the components were given small initial velocities in [1], namely,

$$\dot{x}_1 = +0.00005,\ \dot{y}_1 = 0; \quad \dot{x}_2 = +0.00005,$$
$$\dot{y}_2 = 0; \quad \dot{x}_3 = -0.0001, \quad \dot{x}_3 = 0. \qquad (1)$$

This enabled us to continue the calculations until a true or conditional [1] decay of the triple system occured for 62 out of the 100 randomly chosen initial conditions. For 38 of these conditions the calculations could not be continued right up to decay, since the very close binary approach occurred before the decay of the system.

To overcome this difficulty we have not linearized the equations of motion by the method described in [2]. The equations of motion which depend on the distances between the components are then linear in these distances and, consequently, they remain finite even for close approaches. The program of the calculations was such that, as soon as the distance between any two components became less than 0.01 in the adopted units, the calculation was carried out in terms of the linearized equations. If, on the other hand, all three distances were greater than 0.01, the calculation was based on the usual nonlinear equations. The units of distance, time, and mass were taken to be the same as in [1].

The equations of motion were integrated as in [1] for 100 systems. The initial random configurations were exactly the same as in [1], but the initial velocities of the components were taken to be zero. This means, of course, that we have in fact examined dynamically new triple systems, and the present results are independent of those in [1]. As expected, the motions ended in decay and, in all cases, this occurred after a triple approach of the components.

The results are summarized in Table 1 (see Appendix). The first column of this table gives the running number of the initial conditions in the random-number sequence used in [1]. The second and third columns give the corresponding values of random numbers ξ and η which define the initial configuration of the triple system. The fourth column gives the decay time T of the system in units of the mean time τ of motion of a component through the system. If the decay is conditional, i.e., if one of the components departs to a distance of ten units but the Tevzadze criterion [3] for decay is not satisfied, then the decay time in column 4 is indicated by an asterisk. The fifth column gives the value of the perimeter p of the configurational triangle for the triple approach followed by decay. The values of p are expressed in units of the mean size a of the system. If in a given triple system a close triple approach (with a smaller perimeter of the configurational triangle) has in some way led to decay, the corresponding value of p in column 5 is indicated by an asterisk. The sixth column gives the total energy E (with minus sign) of the triple system. The seventh column gives $-\Delta E/E$, where ΔE is the accumulated error due to numerical integration up to the point of decay. The eighth column gives $-\Delta'E/E$, where $\Delta'E$ is the difference between the energy of a component leaving the system and the energy required by the Tevzadze criterion. In the case of conditional decay, the numbers in this column should, of course, be negative. The ninth and tenth columns give the numbers n_1 and n_2 of triple approaches of the components with $p \leq 1.0\,a$ and $p \leq 0.6\,a$, respectively, which occur prior to decay. Finally, column 11 gives the number of binary approaches k with $r_{i,\,i+1} \leq 0.2\,a$, and column 12 gives the number of transitions s from the nonlinearized to the linearized equations of motion.

As indicated above, this transition was carried out whenever the distance between any two components was less than 0.01. In some cases, s is greater than k; in others it is less. This depends on the value of a, but is also connected with the fact that the number of binary approaches k is counted up to the point of decay,

Table 1

No.	ξ	η	T	p	$-E\cdot10^{-7}$	$\frac{\Delta E}{-E}$	$\frac{\Delta' E^3}{-E}$	n_1	n_2	k	s
1	2	3	4	5	6	7	8	9	10	11	12
1	0.152107965	0.516951804	148.4	0.2313	2.329	0.012	0	3	1	98	51
2	0.022976316	0.521562782	1.2	0.3476	2.376	0	0	1	0	2	0
3	0.269722548	0.476622422	47.1	0.0410	2.116	0.002	3.380	1	1	36	56
4	0.302172674	0.431941366	39.8	0.2584	1.911	0.003	0.098	1	0	25	19
5	0.005310069	0.666154059	224.8*	0.2073	2.918	0.004	−0.088	8	3	74	29
6	0.184999239	0.600560507	15.9	0.2844	2.645	0.001	0	1	0	7	4
7	0.005159629	0.385816848	2.0	0.1392	1.944	0	0.487	1	1	1	1
8	0.157232181	0.393068735	125.6	0.2266*	1.892	0.004	0.356	5	3	62	25
9	0.032952633	0.367271030	132.0	0.2390	1.888	0.003	0	1	0	128	15
10	0.248615885	0.122069321	132.9	0.0823	0.971	0.001	0.824	2	1	90	10
11	0.425790152	0.158446944	147.6	0.1151*	0.558	0.008	0.577	3	2	83	41
12	0.426903973	0.236476070	153.4	0.1312	0.915	0.007	0.552	3	1	15	17
13	0.282590939	0.617363867	324.0	0.2509*	2.707	0.006	0.263	5	2	172	39
14	0.189034578	0.099451113	21.3	0.2080*	1.116	0.005	3.138	2	1	16	23
15	0.008491508	0.593936046	44.3*	0.3484*	2.639	0	−0.065	2	1	3	1
16	0.078425390	0.349994402	31.8	0.0560	1.822	0.903	3.304	1	1	25	22
17	0.115279340	0.749806025	148.4	0.4119	3.260	0.005	0	1	0	76	29
18	0.012014848	0.514189941	1.2	0.2349	2.351	0	1.052	1	1	1	0
19	0.041515318	0.353444473	26.8	0.0947	1.848	0	0.583	1	1	21	6
20	0.355454774	0.287990767	34.6	0.0902	1.243	0.002	0.841	2	2	27	9
21	0.146330588	0.668451531	44.9	0.2226	2.924	0.002	0.408	1	1	21	10
22	0.123048894	0.188874396	120.4	0.1558	1.401	0.002	3.404	1	1	75	24
23	0.093379200	0.033925931	194.4	0.1757	1.281	0.007	0	2	0	8	36
24	0.391574265	0.032756103	104.7	0.0913	0.282	0.002	0.004	2	0	44	58
25	0.263022343	0.247991904	28.2	0.2695	1.267	0.001	0	1	0	15	3
26	0.097623704	0.760908565	516.4	0.3183	3.306	0.013	0.002	5	0	26	70
27	0.165730470	0.139119704	70.6	0.1568	1.233	0.002	0.218	1	1	45	10
28	0.112470028	0.564850334	287.0	0.1009	2.520	0.018	1.804	2	1	150	88
29	0.163015720	0.465528719	171.0	0.1240	2.137	0.002	0.128	4	1	116	56
30	0.453271966	0.242887395	61.2*	0.1920*	0.919	0.010	−0.017	2	1	78	93
31	0.109806163	0.283848681	18.6	0.1673	1.624	0.001	0.276	1	1	15	5
32	0.071236454	0.567962233	109.8	0.2196	2.538	0.002	0.385	1	1	101	70
33	0.048520920	0.290818296	71.5	0.1025	1.687	0.003	0.879	2	1	63	17
34	0.224688133	0.572102054	372.7	0.2223	2.522	0.019	0.103	4	2	239	111
35	0.480236434	0.134853383	464.7	0.0400	0.386	0.006	18.368	5	2	442	336

Table 1 (cont'd.)

1	2	3	4	5	6	7	8	9	10	11	12
36	0.101054627	0.091224803	64.6	0.1181	1.304	0.005	0.906	2	2	45	66
37	0.203387744	0.658870561	103.5	0.3010	2.863	0.006	0	2	0	76	25
38	0.236529953	0.551711569	38.2	0.1704	2.436	0.002	0	1	1	17	7
39	0.470810730	0.135874846	86.0	0.1382	0.397	0.004	1.047	2	0	57	40
40	0.446076458	0.206502066	113.3*	0.1852	0.748	0.011	−0.002	3	0	48	36
41	0.302550914	0.196574135	133.3	0.2406*	1.004	0.017	0	2	0	170	96
42	0.455307350	0.188735628	64.9	0.1081	0.654	0.005	0.664	2	1	40	27
43	0.199431868	0.713511463	6.0	0.4047	3.114	0.003	0	2	0	3	11
44	0.210788097	0.581520233	210.3	0.0561	2.563	0.009	0.900	5	2	234	47
45	0.308297011	0.301011953	12.1*	0.1571*	1.371	0.002	−0.045	6	1	53	14
46	0.221349313	0.127070459	37.1	0.4558	1.065	0.006	3.580	1	1	28	49
47	0.330832116	0.495365951	61.5	0.1862	2.170	0.007	0	1	1	34	31
48	0.046961055	0.558717858	101.5	0.1917	2.507	−0.002	0.171	1	1	103	4
49	0.058727505	0.167417674	22.9	0.3797*	1.446	0.003	0	2	0	43	29
50	0.100298172	0.425877548	9.8*	0.3567	2.037	0.005	−0.008	1	0	7	42
51	0.000427526	0.486750493	1.9	0.1472	2.258	0.001	0	1	1	3	8
52	0.055152381	0.426512804	171.4*	0.1908	2.057	0.004	−0.068	2	1	110	29
53	0.004454218	0.145784306	115.8	0.0687	1.441	0.003	1.110	4	1	49	38
54	0.077480639	0.565092305	181.4	0.3079*	2.527	0.005	0.071	5	0	83	35
55	0.473503305	0.116893831	39.9	0.0824	0.314	0.004	0.559	1	1	22	38
56	0.386569322	0.339951672	276.4	0.2365	1.439	0.005	0	5	2	268	36
57	0.101269911	0.306704186	32.4*	0.2939	1.690	0.008	−0.025	1	0	19	51
58	0.435627483	0.049049590	370.4	0.0884*	0.162	0.008	0.114	6	2	287	283
59	0.176293211	0.167790923	8.7	0.2053	1.259	0.001	0.052	1	0	7	3
60	0.305717285	0.373832423	65.9	0.3163	1.667	0.001	0	1	0	67	14
61	0.437455895	0.150593949	8.1	0.2210	0.504	0.002	0	0	0	0	7
62	0.068169446	0.038072853	23.1	0.0547	1.317	0.003	1.854	1	1	17	33
63	0.159779582	0.581268387	98.3*	0.2754*	2.573	0.010	−0.018	2	0	71	62
64	0.359693815	0.224207350	63.4	0.1326	0.963	0.004	0	3	1	43	18
65	0.306897814	0.264435224	26.9	0.2508	1.234	0.001	0.076	1	0	23	5
66	0.152497417	0.286843464	149.8*	0.2958*	1.580	0.010	−0.033	3	0	59	73
67	0.136803084	0.167749452	100.9	0.0879	1.340	0.005	0.899	2	1	60	27
68	0.258888896	0.567517590	79.1	0.2182	2.496	0.003	0	2	1	41	11
69	0.378927315	0.238608354	166.3	0.2977*	0.991	0.006	0	3	0	105	80
70	0.386784473	0.009493945	11.4	0.1155	0.282	0.005	0	1	0	35	53

Table 1 (cont'd.)

1	2	3	4	5	6	7	8	9	10	11	12
71	0.379442579	0.344937472	101.8	0.1997*	1.468	0.005	1.217	4	0	42	27
72	0.150913059	0.733827112	31.2	0.3051	3.195	0.002	0.250	1	0	15	8
73	0.035567328	0.125720780	101.1	0.2540	1.478	0.005	0	1	0	44	23
74	0.017131688	0.597193411	1.9	0.2695	2.651	0	0.017	1	0	1	0
75	0.171277915	0.163977452	177.0	0.1151	1.263	0.009	0.095	3	2	106	48
76	0.280145301	0.040746517	98.5	0.1203	0.740	0.002	0	2	1	54	52
77	0.111245697	0.032869383	105.1	0.2179	1.250	0.012	0.111	1	0	69	50
78	0.129733170	0.710744272	32.6	0.3629	3.098	0.002	0	2	0	17	10
79	0.177281222	0.520422521	186.6	0.3469	2.332	0.006	0	4	1	97	51
80	0.110426012	0.387555532	10.8*	0.2276	1.911	0	−0.037	1	0	7	3
81	0.242888665	0.198203787	110.8	0.1752*	1.166	0.008	0	5	1	70	40
82	0.142397675	0.385411857	27.1	0.0726	1.880	0.003	2.777	1	1	13	22
83	0.118589788	0.456797180	59.4	0.2351	2.130	0	0	1	0	26	8
84	0.184943235	0.016807856	389.2	0.1698	1.065	0.019	0.154	2	0	332	269
85	0.013188964	0.429651778	2.0	0.2564	2.074	0	0.512	1	0	3	1
86	0.085778906	0.072485138	85.3*	0.2461	1.312	−0.014	−0.057	1	0	51	41
87	0.450134134	0.230877165	5.4	0.1733	0.862	0.003	0.119	1	0	5	12
88	0.432697866	0.311230217	102.2	0.2149*	1.269	0.005	0.070	5	3	60	18
89	0.048364955	0.311528956	41.0	0.1287	1.737	0	0.831	1	1	29	6
90	0.237107399	0.295829168	18.0	0.1376	1.474	0	0.186	2	1	8	5
91	0.199219026	0.217809880	33.9	0.1501	1.319	0	0.328	1	1	25	4
92	0.199254563	0.392218395	122.8	0.1070	1.851	0.003	0.449	2	1	82	17
93	0.196018586	0.471612279	31.8*	0.2474	2.141	0.013	−0.016	1	0	16	47
94	0.144736081	0.248008566	114.9	0.0716	1.492	0.010	1.245	1	1	82	66
95	0.139575010	0.447612270	9.6	0.3226	2.087	0.002	0	1	0	6	10
96	0.468459592	0.180165843	387.8	0.1950*	0.603	0.009	0.061	5	1	299	46
97	0.347295515	0.249211481	88.1	0.2745	1.094	0.002	0.063	0	0	8	78
98	0.174891833	0.530514918	750.2*	0.3447*	2.372	0.014	−0.027	15	2	176	108
99	0.255720500	0.025414176	289.1	0.2074	0.826	0.016	0.032	2	0	354	331
100	0.007783970	0.488557558	2.0	0.1896	2.264	0	0	1	1	10	0

Table 2

	n	$\frac{\Delta' E}{-E}$
$0<p\leqslant 0.1$	15	2.45
$0.1<p\leqslant 0.2$	35	0.412
$0.2<p$	50	0.205

Table 3

No.	T	p	$-E\cdot 10^{-7}$	$\frac{\Delta E}{-E}$	$\frac{\Delta' E}{-E}$	$M\cdot 10^{-4}$	$\frac{\Delta M}{M}\cdot 10^{-4}$	n_1	n_2	k	s
11	5.0*	0.2532	0.483	0.019	—0.024	0.1584	6.6288	0	0	101	16
13	52.5	0.1354	2.632	0.005	0.409	0.6172	— 0.0102	3	2	33	2
14	61.6	0.0249	1.041	0.004	29.241	0.0994	0.0289	2	2	38	10
18	99.2*	0.1920	2.276	0.008	—0.015	0.5142	12.1373	1	1	87	36
19	105.6	0.2113	1.173	0.006	0.498	0.3534	0.0424	1	1	63	32
27	64.4	0.2749	1.158	0.014	0	0.1394	— 0.2360	2	0	50	10
30	118.9	0.1294	0.844	0.012	7.628	0.2429	0.1824	1	1	74	8
36	40.5	0.0378	1.228	0.007	8.371	0.0912	0.2041	1	1	17	63
38	50.0	0.3875	2.361	0.004	0.222	0.5517	— 0.0815	1	0	39	21
40	4.4	0.1616	0.673	0.004	0.135	0.2065	0.1135	6	3	29	9
43	1.2	0.2814	3.039	0 001	0.386	0.7135	— 0.0014	1	0	2	4
45	79.2	0.1496	1.296	0.005	1.384	0.3010	0.0511	2	2	59	27
46	38.6	0.2091	0.990	0.003	0	0.1271	—53.0684	1	0	6	14
49	109.9	0.1562	1.371	0.008	0.737	0.1674	2.729	8	3	63	25
50	68.2	0.4635	1.962	0.003	0.001	0.4259	— 0.4273	0	0	56	11
52	46.1	0.4235*	1.982	0.016	0	0.4205	— 1.8671	2	0	82	69
56	257.9	0.0420	1.364	0.016	0.968	0.3400	— 0.1398	3	1	74	81
57	47.2	0.2286	1.615	0.019	0.188	0.3067	1.3048	1	0	15	4
58	92.1	0.1824	0.087	0.001	0.232	0.0490	0.0369	2	0	26	18
59	208.6	0.1496	1.184	0.006	0.010	0.1678	0.1789	3	2	137	77
62	179.8	0.1128	1.242	0.015	1.246	0.0381	0.2422	2	1	95	70
65	18.7	0.2518	1.159	0.002	0	0.2644	0 0096	1	0	15	9
70	231.5	0.0616	0.208	0.015	0.616	0.0095	2.236	3	1	144	83
74	256.6*	0.3306	2.576	0.013	—0.007	0.5972	0.0214	1	0	154	73
75	129.9	0.0447	1.188	0.011	5.412	0.1640	0.0025	1	1	67	71
79	2.0	0.2484	2.258	0.002	7.139	0.5204	0.0020	1	0	2	8
83	20.6	0.3832*	2.055	0.001	0	0.4568	— 0.0447	2	0	15	9
84	282.3	0.1368	0.990	0.010	0	0.0168	0.1052	2	1	230	160
86	102.4	0.1391	1.237	0.008	0.495	0.0725	— 0.5062	1	1	57	36
87	33.8	0.1308	0.787	0.001	0.162	0.2309	— 0.0039	3	1	23	17
88	83.0*	0.2938*	1.194	0.004	—0.062	0.3112	0.5633	2	0	75	185
89	146.8	0.3290*	1.662	0.002	0.298	0.3115	— 0.0066	3	0	72	45
90	30.8*	0.2742	1.399	0.001	—0.010	0.2958	— 0.0644	1	0	5	7
91	370.2	0.1278	1.244	0.018	0.384	0.2178	18.1630	5	1	268	191
92	31.8	0.2881	1.776	0.003	0.227	0.3922	0.0090	1	0	24	12
95	167.6	0.1147	2.012	0.005	2.485	0.4476	0.3016	1	1	128	68
99	168.3*	0.2276	0.750	0.008	—0.027	0.0254	— 0.0086	2	0	68	48
100	204.7	0.2018*	2.189	0.005	0	0.4886	— 0.1458	1	1	46	24

i.e., the point of triple approach leading to decay, while the number of linearizations is counted to the end of the calculation.

The data given in the table show that, out of the total number of 100 decays, 12 were conditional decays. In 81 cases the decay occurred after the closest triple approach. In 19 systems a closer triple approach occurred prior to the triple approach leading to decay. The number of such systems among those in which a conditional decay was noted was 6, i.e., 50%, while in systems which had undergone true decay, the number was 13, i.e., only 16%. This is clearly reasonable.

As a result of the linearization procedure the accumulated errors were smaller than in [1]. The accumulated error in the principal angular momentum was small both here and in [1]. It was quite negligible in the present work and we shall not cite it.

The uncertainty in energy due to the particular computational method which we used tends to assume positive values and is thus accumulated more rapidly. The data of column 7 shows that the relative energy uncertainty at the time of decay does not exceed 0.02, and in 84 cases is less than 0.01, including 55 cases in which it is less than 0.005 and 13 cases where it is less than 0.0005.

It is obvious that the decay of a triple system is a real phenomenon and is not due to computational factors such as the accumulation of energy errors which, in this method, are always positive. The fact that the decay of the system is not a result of computational errors is also confirmed by comparison of the data in columns 7 and 8. As a rule, the difference between the energy of a component leaving the system and the energy required by the Tevzadze criterion is much greater than the energy error accumulated during calculations. It is also readily seen that $-\Delta'E/E$ is very dependent on the closeness of the triple approach resulting in decay. For small values of p the values of $-\Delta'E/E$ are, as a rule, large. This is indicated in Table 2, where n is the number of triple approaches with given p.

Thus, the decay of triple systems occurs as a result of a triple approach of the components. The energy carried off by the escaping component is statistically greater for closer triple approaches.

The statistical decay parameters of triple systems whose components were at rest at the initial time were calculated from the data of Table 1 and were found to be as follows: mean decay time, $\overline{T} = 110.1\tau$; root-mean-square deviation of the decay time $\sigma = 125.0\tau$.

We can therefore write

$$\overline{T} = (110.1 \pm 12.5)\,\tau. \qquad (2)$$

We recall that τ is the mean time of motion of a component through the system. This quantity is defined rigorously in [1].

As noted above, with the initial conditions adopted in [1], the calculations were not completed in 38 cases out of 100. In the present paper the linearization of the equations of motion has enabled us to fill in this gap. The final results are given in Table 3, where the columns for ξ and η are omitted since they merely repeat the corresponding numbers in Table 1 and in [1]. The values of M and $\Delta M/M$ are given as in [1]. The remaining columns are self-explanatory.

According to Table 3 and the table in [1], we have: mean decay time $\overline{T} = 80.7\ \tau$; root-mean-square deviation $\sigma = 83.0\tau$.

Therefore, in this case, i.e., for triple systems with plane motion, we have

$$\overline{T} = (80.7 \pm 8.4)\,\tau. \qquad (3)$$

The presence of small nonzero velocities [defined by Eq. (1)] at the initial time should not statistically affect the value of $\overline{T}$. The difference between Eqs. (2) and (3) is most likely to be accidental. If we combine them we find that for triple systems with plane motion and small initial velocities

$$\overline{T} = (95.4 \pm 6.9)\,\tau. \qquad (4)$$

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28 September 1967 Leningrad State University